

Supervision 1 - Parameter Estimation

Q1.

A

$$\mu = 12, \sigma = 4, x = 19$$

→ Convert to standard normal distribution - $z = \frac{x - \mu}{\sigma}$

$$z = 1.75$$

$$P(z > 1.75) = 4.01\%$$

B

$$\mu = 9, \sigma = 3, x = 15$$

$$z = 2$$

$$P(z > 2) = 2.3\%$$

This means store A has a higher probability of having weekly sales exceed 19 than store B's probability of weekly sales exceeding 15. So store A is more likely to run out of stock.

Q2. $n = 43$ $\mu = 1.15$ $\sigma = 0.482$

(a) Referring to distribution table, at the 95% confidence level we see a critical z value of 1.96, which I will approximate to 2.

$$1.15 \pm 2 \times \frac{0.482}{\sqrt{43}} = 1.0 \leftarrow \mu \rightarrow 1.30$$

Therefore 95% of the time the true mean would be between 1 and 1.3; this is my confidence interval.

(b) 99% confidence level necessitates a z value of 2.58, which I will approximate to 3.

$$1.15 \pm 3 \times \frac{0.482}{\sqrt{43}} = 0.93 \leftarrow \mu \rightarrow 1.37$$

99% of samples of this nature will have a true mean of between 0.93 and 1.37

Q3. $n = 65$ $\mu = 423$ $\sigma = 115$

(a) $423 \pm \frac{3(115)}{\sqrt{65}} = 380 \leftarrow \mu \rightarrow 465.80$. This is my 99% confidence interval

b) ?

Q4. $n=100$ $q=0.1$

(a) $STEP = \sqrt{\frac{0.1(1-0.1)}{100}} = 3/100$

95% confidence interval requires $z=2$

$$0.1 \pm 2(3/100) = 0.04 \leftarrow p \rightarrow 0.16$$

The width of confidence interval is 0.12 or 12 percentage points.

(ii) $n=400$

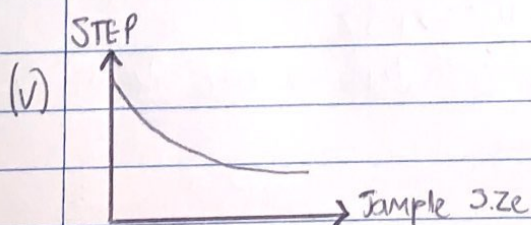
$$0.1 \pm 2\sqrt{0.1(0.9)/400} = 0.07 \leftarrow p \rightarrow 0.13$$

(iii) $n=900$

$$0.1 \pm 2\sqrt{0.1(0.9)/900} = 0.08 \leftarrow p \rightarrow 0.12$$

(iv) $n=1600$

$$0.1 \pm 2\sqrt{0.1(0.9)/1600} = 0.085 \leftarrow p \rightarrow 0.115$$



STEP decreases as sample size increases, but this decreases happens at a decreasing rate as sample size gets larger. This means there are decreasing returns to justify the costs of increasing sample size beyond a certain point.

Q5. $n = 65$ $\bar{x} = 11,918.55$ $s = 12,922.66$

(a) $STEM = 1602.858703$

(b) Confidence interval = $\bar{x} \pm 2STEM$
 $= 11,918.55 \pm 3205.717407$
 $= 8712.83 \leftarrow \mu \rightarrow 15,124.27$

(c) 0.446

(d) $0.446 \pm 2\sqrt{\frac{0.446(1-0.446)}{65}} = 0.32 \leftarrow \mu \rightarrow 0.57$
 proportion of no-cash customers

95% of samples of this type will have a true ~~mean~~ μ between 32% and 57%.

Q6.

(a) $n = 900$ $p = 0.4$ $z = 2$ for 95% CI

$0.4 \pm 2\sqrt{\frac{0.4(0.6)}{900}} = 0.367 \leftarrow \mu \rightarrow 0.433$

Using a 95% confidence level, between 36.7 and 43.3% of passengers would have been affected by the one bag policy.

(b) $500 \times 0.367 = 183.5$
 $500 \times 0.433 = 216$

Between 184 and 216 passengers would have been affected by the policy on a flight of 500 passengers, at the 95% confidence level.

Supervision 1 - Hyp. Testing

$$1. \mu = 210 \quad \sigma = 7.50$$

$$M = 208 \quad n = 100$$

Here, the null hypothesis is that workers in the factory have the same mean wage as the wider industry, and the alternative hypothesis is that the two figures are different. Alternatively:

$$H_0: M = \mu \quad \text{and} \quad H_1: M \neq \mu$$

$$t = \frac{M - \mu}{\text{STEM}} \quad t = \frac{208 - 210}{7.5/10} \quad t = 2.67$$

$$|t| > 1.96$$

At the 5% significance level, the critical value for z is 1.96. If the absolute value of t is greater than this, we reject H_0 . ~~1.96~~ $2.67 > 1.96$ so at the 5% sig level we reject the hypothesis that workers in the factory are paid the industry average.

$$2. \quad p = 0.4 \quad q = 0.5125$$

Assessing at the 1% level requires a 99% confidence level. At this confidence level, the critical z value is given as 2.58.

$$t = \frac{q - p}{\text{STEP}} \quad t = \frac{0.5125 - 0.4}{\sqrt{0.4(0.6)/80}} \quad t = 2.05$$

$$H_0: p = q \quad H_1: p \neq q$$

Applying our conditions for rejecting H_0 , that $|t| > z$, we see that $2.05 < 2.58$, so we cannot reject H_0 . At the 1% significance level, we don't reject the hypothesis that the marketing campaign did not result in a change.

3. $\mu = 10.1$

$n = 260$ $M = 8.6$ $S = 5.7$

$H_0: M = \mu$ $H_1: M \neq \mu$

Null hypothesis is that there is no difference between Pro Bowl tourists and normal tourists.

$$t = \frac{8.6 - 10.1}{5.7 / \sqrt{260}} = -4.24$$

At the 5% significance level, $z = 1.96$. $|t| > z$, so we reject the null hypothesis of no difference between the 2 groups.

4. Weekday

Saturday

(a) $M_1 = 337.84$

$M_2 = 318.45$

$S_1 = 58.86$

$S_2 = 60.98$

$n_1 = 85$

$n_2 = 85$

(b) $\mu = 330$ $M = 337.84$ $S =$

$H_0: \mu = M$

$z = 1.96$ at 5% significance level

$H_1: \mu \neq M$

$t = \frac{M - \mu}{S / \sqrt{N}}$

$t = 7.84 / 58.86 / \sqrt{85}$

$t = 1.228$

$1.23 < 1.96$ so we can't reject the null hypothesis that $\mu = M$, or that the population mean is 330.

(c) $\mu = 330$ $M = 318.45$ $S = 60.98$ $z = 1.96$ for 5% sig. level

$$t = \frac{318.45 - 330}{60.98 / \sqrt{85}}$$

$t = -1.746$

$|t| < z$ so we don't reject the Manager's claim that the population mean is 330, at the 5% significance level

(d) Difference of 2 Means

$$t = (M_1 - M_2) / \text{STEOM}$$

$$n_1 = 85 \quad S_1^2 = 3464.5 \quad M_1 = 337.84$$

$$n_2 = 85 \quad S_2^2 = 3718.56 \quad M_2 = 318.45$$

$$t = \frac{M_1 - M_2}{\sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2} \times \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$t = 2.109271695$$

$$t = 2.11$$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

2.11 > 1.96, So at the 5% significance level we reject the null hypothesis that the population means are the same.